Mathematical Performance Model for DT-HSBP

This document establishes the complete analytical framework for the **Digital Twin–Assisted Hierarchical Secure Broadcast Protocol (DT-HSBP)** in accordance with the supervisor's research checklist, linking the stochastic event processes to closed-form equations for communication cost, computational overhead, efficiency, and scalability.

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1. Stochastic Event Modeling (Checklist Points 2, 3, 4)

1.1 Model Selection & Justification (Poisson Point Process)

In dynamic Unmanned Aerial Vehicle (UAV) swarms, membership events (node arrival/join, voluntary departure/leave, and component or link failure) are driven by independent operational and environmental factors across distributed nodes. The standard and mathematically rigorous model for such memoryless, independent dynamic event arrivals is a **homogeneous Poisson Point Process (PPP)**:

- **Stationary increments:** Event rates remain invariant over operational intervals.
- **Independent increments:** The occurrence of an event in one time interval $[t_1, t_2]$ is independent of occurrences in disjoint intervals.
- **Single events:** The probability of simultaneous overlapping arrivals in an infinitesimal window dt is $o(dt)$.

1.2 Event Rates and Probability Distributions

Let T be the simulation or operational time window. We define three independent Poisson arrival streams:

1. **Join rate:** $\lambda_J > 0$ (events/second)
2. **Leave rate:** $\lambda_L > 0$ (events/second)
3. **Failure rate:** $\lambda_F > 0$ (events/second)

For an observation window Δt, the number of events $N_X(\Delta t)$ of type $X \in \{J, L, F\}$ follows:

$$P(N_X(\Delta t) = k) = \frac{(\lambda_X \Delta t)^k e^{-\lambda_X \Delta t}}{k!}, \quad k \in \{0, 1, 2, \dots\}$$

The combined exogenous event stream $N(t) = N_J(t) + N_L(t) + N_F(t)$ is a merged Poisson process with aggregated rate:

$$\lambda = \lambda_J + \lambda_L + \lambda_F$$

The probability of an event occurrence within time Δt is:

$$P(\text{Event in } \Delta t) = 1 - e^{-\lambda \Delta t}$$

The probability of each distinct event type conditioned on an event occurring is:

$$P(\text{Join} \mid \text{Event}) = \frac{\lambda_J}{\lambda}, \quad P(\text{Leave} \mid \text{Event}) = \frac{\lambda_L}{\lambda}, \quad P(\text{Failure} \mid \text{Event}) = \frac{\lambda_F}{\lambda}$$

> [!IMPORTANT] **Clarification of Event Probabilities vs. Stability Probability:**
>
> - $P(N_X = k)$ models the **arrival probability of exogenous network events** (membership churn).
> - This is strictly separate from the **probability of UAV physical stability**, which is modeled by the Digital Twin state residual $D_i(t) = |\mathbf{e}_i(t)|_2$.

1.3 Statistical Moments (Checklist Point 5)

For each event type $X \in \{J, L, F\}$ over horizon T:

- **Mean (Expected Value):**

$$\mu_X = \mathbb{E}[N_X(T)] = \lambda_X T$$

- **Variance:**

$$\sigma_X^2 = \text{Var}(N_X(T)) = \lambda_X T$$

- **Standard Deviation:**

$$\sigma_X = \text{SD}(N_X(T)) = \sqrt{\lambda_X T}$$

For empirical realizations across R independent repetitions ($r = 1, \ldots, R$):

$$\bar{X} = \frac{1}{R}\sum_{r=1}^{R} X_r, \quad s^2 = \frac{1}{R-1}\sum_{r=1}^{R}(X_r - \bar{X})^2, \quad s = \sqrt{s^2}$$

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2. Mathematical Performance Model (Checklist Points 6, 7, 8, 9, 10, 11)

2.1 Swarm Architecture & Parameters

- Total swarm size: N UAVs
- Number of clusters: K clusters
- Average cluster size: $n_c = \frac{N}{K}$
- Symmetric cryptographic key size: b_k bytes (e.g., 32 bytes for AES-256)
- Message overhead (headers + control metadata): b_{hdr} bytes
- Total packet size per rekey message: $b = b_{\text{hdr}} + b_k = 56$ bytes

2.2 Mathematical Communication Cost Model (C_{comm})

A. Flat-SBP Baseline

In Flat-SBP, every join, leave, or failure triggers a swarm-wide broadcast from the Swarm Leader (SL) to all N UAVs:

- **Join event:** $N + 1$ UAVs receive the new key:

$$M^{J}_{\text{Flat}}(N) = N + 1$$

- **Leave / Failure event (for B departing UAVs):** If departures occur reactively, each requires a swarm broadcast:

$$M^{L}_{\text{Flat}}(N) = \sum_{j=1}^{B} (N - j) \approx B \cdot N$$

- **Communication Cost in Bytes:**

$$C^{\text{Flat}}_{\text{comm}}(N) = b \cdot M_{\text{Flat}}(N)$$

B. HSBP Baseline (Hierarchical Reactive without DT)

In HSBP, rekeying is localized within the affected cluster C_j:

- **Join event:** Only the affected cluster updates:

$$M^{J}_{\text{HSBP}}(N) = n_c + 1 = \frac{N}{K} + 1$$

- **Leave / Failure event (for B departures without predictive batching):** Each departure triggers an independent local rekey:

$$M^{L}_{\text{HSBP}}(N) = \sum_{j=1}^{B} \left(\frac{N}{K} - j \right) \approx B \cdot \frac{N}{K}$$

- **Communication Cost in Bytes:**

$$C^{\text{HSBP}}_{\text{comm}}(N) = b \cdot M_{\text{HSBP}}(N)$$

C. DT-HSBP Proposed Protocol (Predictive & Proactive Batch Rekeying)

In DT-HSBP, the embedded Digital Twin monitors UAV kinematics, residual energy, and link quality:

1. **Instability Detection:** When a UAV u_i experiences physical disturbances or packet loss, its DT residual exceeds the departure threshold $D_i(t) > \theta_{\text{leave}}$.
2. **Hazard-Driven Realization:** The probability that an unstable UAV materializes as a departure within horizon H is governed by the threshold-anchored hazard:

$$\lambda_{\mathrm{DT}}(S) = \lambda_L \exp\left(\frac{S}{\theta_{\text{leave}}} - 1\right), \quad P_{\text{realize}}(S, H) = 1 - \exp\left(-\lambda_{\mathrm{DT}}(S) H\right)$$

3. **Cluster-Local Batch Rekeying:** When a rekey is triggered in cluster C_j, all B departing/unstable UAVs in that cluster are excluded **simultaneously in a single rekey round**:
   - **Join event:** Same as HSBP, with predictive admission filtering:

$$M_{\mathrm{DT}}^{J}(N) = \frac{N}{K} + 1$$

- **Leave / Batch Leave event:** All B realized departures are batched into **1 single cluster broadcast**:

$$M_{\mathrm{DT}}^{L}(N) = \frac{N}{K} - B$$

- **Rekey rounds:** $R_{\mathrm{DT}} = 1$ (compared to $R_{\mathrm{HSBP}} = B$).

$$\boxed{C_{\text{comm}}^{\mathrm{DT}}(N) = b \cdot \left(\frac{N}{K} - B\right)}$$

2.3 Mathematical Computational Cost Model (C_{comp})

Let:

- $T_{\mathrm{DH}}(m)$: computation time to evaluate a sequential Diffie–Hellman chain of length m.
- T_{enc}: symmetric key encryption/generation cost.
- T_{dec}: follower symmetric key decryption cost.

A. Leader Computation Time (C_{leader})

- **Flat-SBP:** Leader must recompute the sequential key across the entire swarm:

$$C_{\text{leader}}^{\text{Flat}} = \mathcal{O}(N) = N \cdot T_{\mathrm{DH}}$$

- **HSBP:** Only the Cluster Head (CH) recomputes the chain, but must do so for each of the B sequential departures:

$$C_{\text{leader}}^{\text{HSBP}} = B \cdot \mathcal{O}(n_c) = B \cdot \frac{N}{K} \cdot T_{\mathrm{DH}}$$

- **DT-HSBP:** The CH recomputes the chain **once** for the remaining $(n_c - B)$ stable members:

$$C_{\text{leader}}^{\mathrm{DT}} = \mathcal{O}(n_c - B) = \left(\frac{N}{K} - B\right) \cdot T_{\mathrm{DH}}$$

B. Follower Computation Time (C_{follower})

- **Flat-SBP:** All N followers decrypt and update the key for each departure:

$$C_{\text{follower}}^{\text{Flat}} = B \cdot N \cdot T_{\text{dec}}$$

- **HSBP:** Followers in the affected cluster update B times:

$$C_{\text{follower}}^{\text{HSBP}} = B \cdot \frac{N}{K} \cdot T_{\text{dec}}$$

- **DT-HSBP:** Followers in the affected cluster update **once**:

$$C_{\text{follower}}^{\text{DT}} = \left(\frac{N}{K} - B \right) \cdot T_{\text{dec}}$$

2.4 Communication Efficiency & Performance Gain (Checklist Points 9, 10)

We define the Communication Efficiency η as the ratio of useful key dissemination payload to total transmitted bytes:

$$\eta = \frac{b_k}{b} \cdot \frac{1}{\text{Rekeys}}$$

The Performance Gain $G(N)$ of DT-HSBP relative to a baseline protocol $\mathcal{B} \in \{\text{Flat-SBP}, \text{HSBP}\}$ is defined as:

$$G_{\text{comm}}^{\mathcal{B}}(N) = \frac{C_{\text{comm}}^{\mathcal{B}}(N) - C_{\text{comm}}^{\text{DT}}(N)}{C_{\text{comm}}^{\mathcal{B}}(N)} \times 100\%$$

Gain against Flat-SBP:

$$G_{\text{comm}}^{\text{Flat}}(N) = \frac{B \cdot N - \left(\frac{N}{K} - B\right)}{B \cdot N} \times 100\% = \left(1 - \frac{1}{BK} + \frac{1}{N}\right) \times 100\%$$

For $K = 5$ clusters and $B = 3$ batch departures:

$$G_{\text{comm}}^{\text{Flat}}(N) \approx \left(1 - \frac{1}{15}\right) \times 100\% = \mathbf{93.33\%}$$

Gain against HSBP:

$$G_{\text{comm}}^{\text{HSBP}}(N) = \frac{B \cdot \frac{N}{K} - \left(\frac{N}{K} - B\right)}{B \cdot \frac{N}{K}} \times 100\% = \left(1 - \frac{1}{B} + \frac{K}{N}\right) \times 100\%$$

For $B = 3$ departures in a cluster:

$$G_{\text{comm}}^{\text{HSBP}}(N) \approx \left(1 - \frac{1}{3}\right) \times 100\% = \mathbf{66.67\%}$$

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3. Mathematical Justification of DT Overhead Reduction (Checklist Point 13)

Why does the Digital Twin mathematically reduce rekeying overhead?

1. **Elimination of Desynchronization Recovery Penalty:** In reactive protocols (Flat-SBP, HSBP), an unstable UAV (experiencing link degradation or imminent power failure) is included in the rekey broadcast. When it fails to receive the rekey message, the Diffie-Hellman chain or group key breaks, triggering a costly **recovery rekey protocol** with communication penalty $C_{\text{recovery}} = \mathcal{O}(N)$ and latency timeout T_{timeout}. Under DT-HSBP:

$$P_{\text{desync}}^{\text{DT}} = P\left(D_i(t) \leq \theta_{\text{leave}} \mid \text{UAV fails in } H\right) \to 0$$

Unstable UAVs are proactively excluded **before** cryptographic synchronization, eliminating recovery cycles.

2. **Batch Amortization Factor:** When B unstable nodes are detected within the DT prediction window H, DT-HSBP amortizes the rekeying cost across all B departures into a single cryptographic transition. The rekey reduction factor is exactly:

$$\mathcal{R}_{\text{factor}} = \frac{R_{\text{DT}}}{R_{\text{HSBP}}} = \frac{1}{B}$$

This reduces the number of rekeys by $\left(1 - \frac{1}{B}\right) \times 100\%$.

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4. Scalability Analysis (Checklist Point 14)

As the swarm size $N \to \infty$ with a fixed number of clusters K, or when $K = \mathcal{O}(\sqrt{N})$:

Metric	Flat-SBP	HSBP	DT-HSBP	DT-HSBP Scalability Impact
Leader Comp. Time	$\mathcal{O}(N)$	$\mathcal{O}(N/K)$	$\mathcal{O}(N/K)$	**Near-constant per cluster**
Follower Comp. Time	$\mathcal{O}(N)$	$\mathcal{O}(N/K)$	$\mathcal{O}(N/K)$	**Independent of non-affected clusters**
Message Overhead (Join)	$N+1$	$N/K+1$	$N/K+1$	**Localized by factor of** $1/K$

Metric	Flat-SBP	HSBP	DT-HSBP	DT-HSBP Scalability Impact
Message Overhead (Leave)	$B \cdot N$	$B \cdot (N/K)$	$\frac{N}{K} - B$	**Amortized by factor of** $\frac{1}{BK}$
SL Communication Overhead	$\mathcal{O}(N)$	$\mathcal{O}(K)$	$\mathcal{O}(K)$	**Strictly** $\mathcal{O}(K)$**, invariant to** N

5. Multi-Scale Experimental Results & Model Validation (Checklist Points 5, 12)

The Stage E Full Join/Leave Study was executed across the five established swarm sizes $N \in [1000, 2000, 3000, 4000, 5000]$ with 3 repetitions per configuration (90 total runs) under paired Poisson event workloads.

5.1 Join Event Numerical Performance (Single Drone Join)

Swarm Size (N)	Protocol	EvtMean ($\bar{N}_J$)	Leader Time (ms)	Follower Time (ms)	Messages Sent ($\bar{M}$)	Bytes Transmitted ($\bar{C}_{\text{comm}}$)	Rekeys
1000	Flat-SBP	59.0	5.83 ± 0.27	0.062 ± 0.047	1001.0 ± 0.0	$56,056 \pm 0$	1.0 ± 0.0
	HSBP	59.0	1.17 ± 0.02	0.024 ± 0.028	201.0 ± 0.0	$11,256 \pm 0$	1.0 ± 0.0
	DT-HSBP	59.0	1.19 ± 0.04	0.026 ± 0.028	201.0 ± 0.0	$11,256 \pm 0$	1.0 ± 0.0
2000	Flat-SBP	59.0	11.75 ± 0.13	0.085 ± 0.004	2001.0 ± 0.0	$112,056 \pm 0$	1.0 ± 0.0
	HSBP	59.0	2.35 ± 0.10	0.015 ± 0.002	401.0 ± 0.0	$22,456 \pm 0$	1.0 ± 0.0
	DT-HSBP	59.0	2.38 ± 0.12	0.016 ± 0.002	401.0 ± 0.0	$22,456 \pm 0$	1.0 ± 0.0
3000	Flat-SBP	59.0	17.72 ± 0.66	0.133 ± 0.012	3001.0 ± 0.0	$168,056 \pm 0$	1.0 ± 0.0
	HSBP	59.0	3.50 ± 0.04	0.023 ± 0.003	601.0 ± 0.0	$33,656 \pm 0$	1.0 ± 0.0
	DT-HSBP	59.0	3.47 ± 0.09	0.022 ± 0.002	601.0 ± 0.0	$33,656 \pm 0$	1.0 ± 0.0

Swarm Size (N)	Protocol	EvtMean ($\bar{N}_J$)	Leader Time (ms)	Follower Time (ms)	Messages Sent ($\bar{M}$)	Bytes Transmitted ($\bar{C}_{\text{comm}}$)	Rekeys
4000	Flat-SBP	59.0	22.36 ± 0.38	0.153 ± 0.016	4001.0 ± 0.0	$224,056 \pm 0$	1.0 ± 0.0
	HSBP	59.0	4.44 ± 0.03	0.024 ± 0.001	801.0 ± 0.0	$44,856 \pm 0$	1.0 ± 0.0
	DT-HSBP	59.0	4.40 ± 0.05	0.023 ± 0.001	801.0 ± 0.0	$44,856 \pm 0$	1.0 ± 0.0
5000	Flat-SBP	59.0	27.88 ± 0.08	0.201 ± 0.005	5001.0 ± 0.0	$280,056 \pm 0$	1.0 ± 0.0
	HSBP	59.0	5.62 ± 0.08	0.031 ± 0.002	1001.0 ± 0.0	$56,056 \pm 0$	1.0 ± 0.0
	DT-HSBP	59.0	5.57 ± 0.07	0.031 ± 0.001	1001.0 ± 0.0	$56,056 \pm 0$	1.0 ± 0.0

5.2 Leave Event Numerical Performance (Batch & Predictive Leave)

Swarm Size (N)	Protocol	EvtMean ($\bar{N}_L$)	Leader Time (ms)	Follower Time (ms)	Messages Sent ($\bar{M}$)	Message Overhead (KB)	Rekeys per Ep.	DT-HSBP Comm Gain
1000	Flat-SBP	37.0	64.12 ± 2.05	27.48 ± 0.88	2903.9 ± 6.7	158.81 ± 0.37	2.9 ± 0.0	93.21% **vs Flat**
	HSBP	37.0	18.39 ± 0.43	7.88 ± 0.18	576.2 ± 1.3	31.51 ± 0.07	2.9 ± 0.0	65.79% **vs HSBP**
	DT-HSBP	37.0	13.25 ± 0.18	5.68 ± 0.08	197.1 ± 0.0	10.78 ± 0.00	1.0 ± 0.0	**(Baseline)**
2000	Flat-SBP	37.0	126.32 ± 1.97	54.14 ± 0.84	5850.2 ± 19.9	319.93 ± 1.09	2.9 ± 0.0	93.21% **vs Flat**
	HSBP	37.0	36.96 ± 0.84	15.84 ± 0.36	1165.4 ± 4.0	63.73 ± 0.22	2.9 ± 0.0	65.93% **vs HSBP**
	DT-HSBP	37.0	26.34 ± 0.38	11.29 ± 0.16	397.1 ± 0.0	21.71 ± 0.00	1.0 ± 0.0	**(Baseline)**
3000	Flat-SBP	37.0	191.15 ± 13.23	81.92 ± 5.67	8567.4 ± 45.9	468.53 ± 2.51	2.9 ± 0.0	93.03% **vs Flat**

Swarm Size (N)	Protocol	EvtMean ($\bar{N}_L$)	Leader Time (ms)	Follower Time (ms)	Messages Sent ($\bar{M}$)	Message Overhead (KB)	Rekeys per Ep.	DT-HSBP Comm Gain
	HSBP	37.0	57.30 ± 5.83	24.56 ± 2.50	1709.0 ± 9.1	93.46 ± 0.50	2.9 ± 0.0	65.06% **vs HSBP**
	DT-HSBP	37.0	39.87 ± 2.05	17.09 ± 0.88	597.1 ± 0.0	32.66 ± 0.00	1.0 ± 0.0	**(Baseline)**
4000	Flat-SBP	37.0	253.55 ± 1.99	108.67 ± 0.85	11565.6 ± 36.4	632.50 ± 1.99	2.9 ± 0.0	93.11% **vs Flat**
	HSBP	37.0	76.01 ± 1.74	32.58 ± 0.75	2308.6 ± 7.2	126.25 ± 0.40	2.9 ± 0.0	65.47% **vs HSBP**
	DT-HSBP	37.0	55.10 ± 4.78	23.61 ± 2.05	797.1 ± 0.0	43.59 ± 0.00	1.0 ± 0.0	**(Baseline)**
5000	Flat-SBP	37.0	365.49 ± 47.87	156.64 ± 20.52	14182.7 ± 19.5	775.62 ± 1.07	2.8 ± 0.0	92.97% **vs Flat**
	HSBP	37.0	101.97 ± 4.81	43.70 ± 2.06	2832.1 ± 3.9	154.88 ± 0.21	2.8 ± 0.0	64.79% **vs HSBP**
	DT-HSBP	37.0	68.95 ± 4.67	29.55 ± 2.00	997.2 ± 0.0	54.53 ± 0.00	1.0 ± 0.0	**(Baseline)**

6. Publication Figures Generated

All figures have been rendered at 300 DPI and saved in figures/ and the artifacts directory:

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![Join Leader Execution Time](C:/Users/Owner/.gemini/antigravity-ide/brain/08e7439a-0e8c-4e8c-9a76-e0adbb1f8f8f/Join
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![Join Follower Execution Time](C:/Users/Owner/.gemini/antigravity-ide/brain/08e7439a-0e8c-4e8c-9a76-e0adbb1f8f8f/Jo
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![Join Message Overhead](C:/Users/Owner/.gemini/antigravity-ide/brain/08e7439a-0e8c-4e8c-9a76-e0adbb1f8f8f/Join_Mess
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![Leave Rekey Activity](C:/Users/Owner/.gemini/antigravity-ide/brain/08e7439a-0e8c-4e8c-9a76-e0adbb1f8f8f/Leave_Reke
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![Communication Performance Gain](C:/Users/Owner/.gemini/antigravity-ide/brain/08e7439a-0e8c-4e8c-9a76-e0adbb1f8f8f/
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